ORGANIC INTELLIGENCE PROTOCOL

Enterprise Business Case, Total Cost of Ownership and ROI Evaluation Framework

A Customer-Specific Method for Measuring Operational Cost, Financial Benefit, Risk and Return

Document Reference
OIP-ENT-FIN-2026-V1

Protocol Reference
OIP v1.0.40 Research-Stage Technical Master Reference

Intended Audience
Chief Executive Officers, Chief Financial Officers, Chief Technology Officers, Risk Officers and Board Audit Committees

Core Principle
Evidence must authorize measurement.

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1 Executive Summary

This document provides a financial and operational evaluation framework for enterprises considering whether the Organic Intelligence Protocol (OIP) may be appropriate for a specific business workflow.

Its purpose is not to promise financial returns. It establishes a reproducible method for measuring costs, benefits, risks and uncertainty using company-specific evidence.

OIP is a research-stage framework. Its broader mathematical and operational methodology contains unresolved definitions and approval requirements. Available synthetic software verification evidence must not be interpreted as proof of empirical, predictive, causal or commercial effectiveness.

An enterprise should distinguish between three questions:

1. Can the proposed system be implemented and operated safely within the company's technical environment?
2. Can its effects on the selected workflow be measured against a credible baseline?
3. Do the measured financial and operational outcomes justify the total cost and residual risk?

The evaluation must permit four possible conclusions:

- **Positive:** Verified benefits exceed relevant costs and enterprise acceptance requirements are satisfied.
- **Neutral:** No material positive or negative effect has been established.
- **Negative:** Measured costs or adverse effects exceed verified benefits.
- **Not Measurable:** Available evidence is insufficient to calculate a reliable result.

A positive financial result does not independently establish scientific validity, regulatory compliance or authorization for unrestricted deployment.

2 Evaluation Principles and Evidence Boundaries

The governing sequence is:

$$\text{Evidence} \rightarrow \text{Measurement} \rightarrow \text{Score} \rightarrow \text{Evaluation} \rightarrow \text{Claim} \tag{1}$$

The central data rule is:

$$\text{Missing} \neq 0 \tag{2}$$

Missing, contradictory, unverified or unavailable information must not be silently converted into a numerical value.

The financial evaluation must follow these principles:

- Use company-specific and traceable financial records.

- Define the baseline before measuring the proposed system.
- Separate measured results from assumptions and projections.
- Do not count the same financial benefit more than once.
- Do not treat time saved as cash saved unless the financial effect is demonstrated.
- Do not treat revenue as profit.
- Do not assign financial value to non-financial benefits without a defensible valuation method.
- Report negative, neutral and unresolved outcomes.
- Preserve the distinction between a software test result and an enterprise outcome.

3 OIP Overview and Current Evidence Status

OIP is a research-stage framework intended to examine evidence, measurement boundaries, decision authorization and system governance.

The available reference material describes mathematical components, boundary rules, audit records and a limited synthetic software reference implementation.

The available evidence does not establish:

- Independent empirical validation across real-world enterprise workflows.
- Predictive or causal effectiveness.
- Universal applicability across industries or datasets.
- Commercial performance or guaranteed financial returns.
- Production readiness or unrestricted deployment authorization.
- Regulatory certification or automatic legal compliance.
- Demonstrated superiority over specific alternative systems.

The synthetic software reference implementation demonstrates selected authorization decision rules. It must not be described as a complete implementation of the broader OIP methodology.

3.1 Decision State Terminology

The reference implementation includes states such as:

- `APPROVED`
- `PENDING_EVIDENCE`
- `BLOCKED`
- `INVALID_INPUT`

These states must be interpreted according to the applicable implementation and its documented decision rules.

An enterprise must not assume that a state automatically authorizes a particular business action. The operational meaning, escalation path and fallback behavior must be specified and tested for the selected use case.

4 Enterprise Baseline and Cost of Inaction

Before evaluating OIP, an enterprise must document the existing workflow.

The baseline should identify:

- Annual case volume and case distribution.
- Current processing time and labour requirements.
- Existing review and escalation costs.
- Verified financial losses.
- Rework and reconciliation costs.
- Contractual service penalties, where applicable.
- Existing technology and infrastructure costs.
- Current error rates based on independently established outcomes.

The Cost of Inaction is the verified financial cost of continuing the existing workflow over a defined period.

It must not include hypothetical losses, unverified opportunity costs or historical losses that the proposed system could not reasonably prevent.

A general baseline cost model is:

$$C_B = C_{\text{labour}} + C_{\text{verified losses}} + C_{\text{rework}} + C_{\text{other baseline costs}} \tag{3}$$

Where:

- C_B is the total baseline workflow cost.
- C_{labour} is the measured labour cost.
- $C_{\text{verified losses}}$ includes only supported and attributable losses.
- C_{rework} is the measured cost of correction and reconciliation.

The categories must be mutually exclusive or adjusted to prevent double-counting.

5 Data Quality and Measurement Readiness

Data quality must be evaluated before financial performance is attributed to a proposed system.

Relevant dimensions include:

- Completeness

- Accuracy
- Validity
- Consistency
- Uniqueness
- Timeliness
- Provenance
- Representativeness
- Missingness
- Contradiction rate
- Independent ground-truth coverage
- Outcome reliability

The enterprise must define acceptance thresholds before evaluation begins.

No universal data quality score or threshold is established by this document.

A dataset can be complete without being accurate. Data quality alone does not establish construct validity, outcome validity or causal attribution.

If the available evidence cannot support a reliable measurement, the result must be reported as:

$$\text{NOT CALCULABLE} \tag{4}$$

6 Total Cost of Ownership

Total Cost of Ownership (TCO) includes the costs required to introduce, operate, maintain and eventually discontinue the proposed system.

Relevant cost categories include:

- Initial integration and engineering.
- Data preparation and mapping.
- Infrastructure and compute.
- Security and privacy assessment.
- Legal and regulatory review.
- Staff training and change management.
- Human review and escalation.
- Monitoring and maintenance.
- Support and incident response.
- Backup and disaster recovery.

- Upgrades and dependency management.
- Decommissioning and exit costs.

A discounted TCO model is:

$$TCO = C_0 + \sum_{t=1}^{Y} \frac{C_t}{(1+r)^t} \tag{5}$$

Where:

- C_0 is the initial cost at the beginning of the evaluation period.
- C_t is the applicable cost during period t.
- Y is the evaluation horizon.
- r is the selected discount rate.

The enterprise must define cost categories, timing, currency and discount rate.

The same cost must not be included both in the initial investment and in recurring operating costs.

7 OIP Mathematical Reference and Boundaries

The following expressions are included as mathematical references from OIP documentation. Their presence in this business evaluation framework does not establish that all variables have complete operational definitions or that the formulas are approved for universal empirical use.

7.1 Infrastructure Independence

$$II = 1 - IDS \tag{6}$$

7.2 Stability Expression

$$Stability = GFL \times II \times AML \tag{7}$$

Equivalently:

$$Stability = GFL \times (1 - IDS) \times AML \tag{8}$$

7.3 LIV Boundary Rule

$$LIV = \frac{RI \times SR \times AR}{IDS} \tag{9}$$

At the boundary:

$$IDS = 0 \Rightarrow LIV = \text{UNDEFINED} \tag{10}$$

No epsilon floor or denominator substitution is permitted for this boundary rule.

This LIV boundary must not be confused with the definition of Infrastructure Independence.

7.4 Score Reference

$$Score = 0.6OE + 0.3CV + 0.1VIS \tag{11}$$

7.5 SIS Reference

$$SIS = 0.5RC + 0.3AFL + 0.2CAS \tag{12}$$

The operational definitions, calibration requirements and approval status of unresolved variables must be checked against the applicable OIP source documents.

These expressions are not financial ROI formulas. They must not be substituted for enterprise cash-flow calculations.

8 OIP-Assisted Workflow Evaluation

An enterprise evaluating an OIP-assisted workflow must document the actual process used during the evaluation.

The documentation should identify:

- Which system receives the input.
- Which validation and authorization rules are applied.
- What each decision state means in the implementation.
- Which cases require human review.
- Who is authorized to approve or override a decision.
- What happens when the system is unavailable.
- How errors, delays and adverse outcomes are recorded.
- How decisions are traced and independently reviewed.

The workflow must not assume that all pending or blocked cases are harmful, or that all approved cases are correct.

Human review cost must be measured from actual review activity:

$$C_{\text{review}} = \sum_{i=1}^{n} H_i \times R_i \tag{13}$$

Where:

- H_i is the actual review time for category i.
- R_i is the applicable fully loaded labour rate.

Time savings should not be treated as cash savings unless they reduce actual expenditure or create measurable additional productive capacity.

9 Independent Ground Truth and Error Evaluation

The enterprise must define the correct outcome independently of the OIP decision state.

A decision state is not itself ground truth.

For example, a blocked case may be correctly blocked or incorrectly blocked. An approved case may be correct or harmful. A pending case may remain unresolved.

The evaluation should distinguish:

- Correct approvals.
- Incorrect approvals.
- Correct blocks.
- Incorrect blocks.
- Unresolved cases.
- Invalid inputs.
- Cases requiring additional evidence.

Formal statistical Type I and Type II errors require a clearly defined hypothesis-testing framework.

Operational false approvals and false blocks should not automatically be labelled as formal Type I and Type II errors.

Independent outcome adjudication, consistent case definitions and documented evaluation rules are required before calculating error rates.

10 Cost of Delay

Delay costs must be measured against the baseline workflow.

Only the incremental delay attributable to the evaluated system should be included.

A general model is:

$$C_{\text{delay}} = C_{\text{incremental service impact}} + C_{\text{incremental abandonment}} \tag{14}$$

The enterprise should identify:

- Baseline processing time.
- Processing time under the evaluated workflow.
- Additional delay attributable to the evaluated system.
- Contractual or operational consequences of delay.
- Independently measured abandonment or cancellation.
- Any overlap with other recorded financial losses.

A pending case is not automatically a financial loss. A delay is not automatically caused by OIP. The enterprise must establish attribution before assigning a financial value.

11 Financial Evaluation and Return on Investment

The financial model must compare equivalent workflows over the same period and at comparable case volumes and case mixes.

Define:

- C_B: Comparable baseline workflow cost.
- C_A: OIP-assisted workflow operating cost, excluding incremental OIP-specific investment K.
- K: Incremental OIP-specific costs.
- S_A: Verified and attributable operational cost reduction.
- G_A: Separately verified additional financial gains.
- L_A: Separately measured additional losses or adverse financial effects.

The comparable operational cost reduction is:

$$S_A = C_B - C_A \tag{15}$$

This comparison is valid only when the baseline and assisted workflows have comparable scope, volume, case mix, time period and accounting boundaries.

If those conditions are not satisfied, the difference must not be reported as an attributable saving.

The net financial benefit is:

$$NB = S_A + G_A - K - L_A \tag{16}$$

Where the denominator is valid and greater than zero:

$$ROI = \frac{NB}{K} \times 100 \tag{17}$$

If K is zero, undefined or unavailable, ROI must be reported as not calculable under this formula.

The cost register must define exactly which expenses are included in C_A and which incremental costs are included in K. Each cost must appear once only.

11.1 Cost per Completed Case

$$C_{\text{case}} = \frac{C_{\text{total comparable workflow}}}{N_{\text{completed cases}}} \tag{18}$$

The numerator and denominator must refer to the same period, workflow scope and case population.

11.2 Net Present Value

$$NPV = \sum_{t=0}^{Y} \frac{\Delta CF_t}{(1+r)^t} \tag{19}$$

Where:

- ΔCF_t is the incremental net cash flow during period t.
- r is the selected discount rate.
- Y is the evaluation horizon.

Initial investment must be included in the period-zero cash flow. It must not be deducted a second time.

The enterprise should document the treatment of taxes, inflation, working capital, cash-flow timing and any terminal value.

NPV should be the primary discounted financial measure.

Internal Rate of Return may be reported where meaningful. Multiple or undefined IRR results must not be concealed.

11.3 Payback Period

Payback should be calculated from cumulative incremental cash flow.

A simple average monthly saving is not sufficient when benefits vary over time, implementation is phased or cash flows are delayed.

12 Scenario and Sensitivity Analysis

The enterprise should prepare conservative, base and favourable scenarios using documented inputs.

Each scenario should identify:

- Case volume.
- Case mix.
- Labour rates.
- Review rates and review time.
- Infrastructure and operating costs.
- Error rates supported by independent outcomes.
- Verified financial losses.
- Implementation and transition costs.
- Delay and service impacts.
- Uncertainty and data limitations.

Scenario values must come from observed company data, documented quotations, defensible assumptions or clearly identified external evidence.

Probabilities must not be assigned to scenarios without a defensible basis.

Monte Carlo simulation should be used only when input distributions and dependencies can be justified.

If a key input cannot be supported, the affected result must be identified as uncertain or not calculable.

13 Controlled Evaluation and Pilot Design

The duration and sample size of an enterprise evaluation must be determined by the selected use case.

Relevant considerations include:

- Case volume and event frequency.
- Expected effect size.
- Statistical power and minimum detectable effect.
- Availability of independent ground truth.
- Seasonality and operational variation.
- Risk of selection bias and contamination.
- Human review capacity.
- Potential harm from incorrect decisions.

Where feasible and appropriate, randomized comparison may be considered.

Where randomization is not practical, a suitable observational or quasi-experimental design may be evaluated.

The design, outcomes, exclusions and acceptance criteria should be documented before results are examined.

A pilot may begin in shadow mode where the enterprise can safely observe decisions without allowing them to control live business outcomes.

A successful pilot does not automatically authorize production deployment.

Separate technical, operational, security, legal and governance approvals may be required.

14 Security, Infrastructure and Operational Resilience

Deployment options should be evaluated using the enterprise's actual technical requirements and cost quotations.

Possible deployment environments include:

- Public cloud.

- Private cloud.
- On-premise infrastructure.
- Hybrid infrastructure.

The assessment should cover:

- Data residency and access control.
- Identity and permission management.
- Encryption in transit and at rest.
- Audit logging and integrity protection.
- Backup and disaster recovery.
- Dependency and supply-chain risks.
- Vendor and infrastructure dependency.
- Incident response.
- Rollback and exit procedures.

Cryptographic hashes can help identify changes to recorded data. A hash alone does not establish confidentiality, authenticity, authorization or complete audit integrity.

Specific security controls must be selected according to the enterprise threat model and verified implementation.

Fallback behavior must be designed for the relevant operational risks. It may involve manual review, suspension, rollback or another documented procedure.

No single fallback behavior should be assumed appropriate for every use case.

15 Human Factors and Organizational Readiness

Financial and operational evaluation must include the effect of the system on people and organizational processes.

Relevant factors include:

- Reviewer workload.
- Training requirements.
- Alert fatigue.
- Automation bias.
- Confirmation and anchoring bias.
- Over-reliance on system outputs.
- Loss of independent judgement.
- Override and escalation behaviour.

- Accountability for final decisions.
- Employee adoption and process change.

Human review should not be treated as a cost-free fallback.

The enterprise should measure actual review time, workload, disagreement rates, escalation outcomes and the quality of independent judgement.

The objective should be appropriate trust and accountable decision-making, not unquestioned acceptance of system outputs.

16 Governance and External Evaluation References

An enterprise may use established external frameworks and methods to structure its own evaluation.

Examples include:

- NIST Artificial Intelligence Risk Management Framework.
- NIST Cybersecurity Framework.
- ISO/IEC 42001.
- ISO/IEC 23894.
- ISO/IEC 27001.
- ISO/IEC 27701.
- Failure Mode and Effects Analysis.
- Threat modelling and system dependency analysis.
- Independent testing and outcome adjudication.

These are external references. Their inclusion does not establish that OIP is certified, compliant or automatically aligned with any particular standard.

Regulatory applicability depends on jurisdiction, sector, system purpose and actual deployment.

Legal and compliance conclusions must be made by appropriately qualified professionals using applicable current requirements.

The enterprise should maintain a register separating:

- Legal requirements.
- Proposed technical controls.
- Verified implementation evidence.
- Unresolved compliance gaps.
- Responsible decision owners.

17 Status Quo, Internal Development and External Evaluation

An enterprise should compare the proposed OIP approach with its current workflow and other available alternatives.

The comparison should use verified evidence rather than assumed advantages.

Relevant dimensions include:

- Total implementation cost.
- Total operating cost.
- Integration complexity.
- Technical feasibility.
- Evidence traceability.
- Data governance.
- Security and resilience.
- Human review requirements.
- Maintenance and support.
- Vendor dependency.
- Exit and migration cost.

No option should be described as inherently cheaper, safer, more accurate or more auditable without evidence from the specific alternatives being compared.

The status quo may remain the preferred option if the proposed change cannot demonstrate sufficient value or acceptable risk.

18 Executive Decision and No-Go Criteria

An enterprise should define its acceptance and rejection criteria before deployment.

Potential reasons to stop, revise or reject an evaluation include:

- Insufficient evidence to measure outcomes.
- Unresolved critical data quality problems.
- Unacceptable residual operational risk.
- Failure of a critical security control.
- Inability to establish independent ground truth.
- Unacceptable human review workload.
- Negative financial outcomes.
- Failure to meet the enterprise's approved financial hurdle.
- Unresolved legal or regulatory restrictions.

- Inability to maintain accountability and oversight.

A negative NPV or ROI should be interpreted using the enterprise's approved financial criteria and evaluation horizon.

The enterprise may decide to reject, revise, suspend, continue in shadow mode or conduct further testing.

No positive result should be assumed in advance.

19 Reporting, Auditability and Reproducibility

The final enterprise report should preserve enough information for an independent reviewer to reproduce the evaluation.

The report should contain:

- Evaluation scope and business purpose.
- Baseline definition.
- Data sources and provenance.
- Data quality findings.
- Measurement definitions.
- Cost assumptions and supporting records.
- Implementation configuration.
- Test and evaluation procedures.
- Independent outcome records.
- Financial calculations.
- Uncertainty and limitations.
- Negative and unresolved findings.
- Approval and decision records.

Every material claim should be traceable to its source, measurement or explicitly stated assumption.

A reported software test result must identify its test scope and evidence status. It must not be presented as proof of enterprise effectiveness.

20 Conclusion

The business value of OIP cannot be established through general claims, assumed savings or illustrative financial projections.

It must be evaluated within a defined enterprise workflow using comparable baseline data, independently measured outcomes, complete cost accounting and documented uncertainty.

The framework in this document gives enterprises a method to calculate their own Total Cost of Ownership, incremental financial benefit, Net Present Value and Return on Investment.

It also allows an enterprise to conclude that the available evidence does not justify adoption.

A positive financial result is only one part of the decision. Technical feasibility, operational safety, governance, legal applicability and independent validation remain separate requirements.

No evidence, no operationalization.

A Variable Dictionary

Variable	Definition
C_B	Comparable baseline workflow cost
C_A	OIP-assisted workflow operating cost, excluding incremental investment K
K	Incremental OIP-specific cost
S_A	Verified attributable operational cost reduction
G_A	Verified additional financial gain
L_A	Additional measured financial loss or adverse effect
NB	Net financial benefit
TCO	Total Cost of Ownership
NPV	Net Present Value
ROI	Return on Investment
r	Discount rate
Y	Evaluation horizon
IDS	Infrastructure Dependency Score
II	Infrastructure Independence
GFL	Grounded Feedback Loop
AML	Adaptive Moral Logic
LIV	OIP variable subject to its documented boundary rule

B Enterprise Data Collection Checklist

Before calculating financial outcomes, the enterprise should collect:

1. Baseline case volume and case mix.
2. Baseline processing and review time.
3. Fully loaded labour rates.
4. Existing technology and infrastructure costs.
5. Verified error and loss records.

6. Rework and reconciliation costs.
7. Independent outcome records.
8. Proposed integration and implementation quotations.
9. Compute and infrastructure costs.
10. Human review and escalation costs.
11. Security, privacy and legal assessment costs.
12. Maintenance, support and exit costs.
13. Evaluation horizon and discount rate.
14. Pre-registered acceptance criteria.

If a required input is unavailable, the enterprise should identify the resulting limitation rather than substitute an unsupported estimate.

C Document Status

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Purpose: To support customer-specific financial and operational evaluation using independently verifiable evidence.

This document does not amend the OIP protocol, approve unresolved mathematical definitions, authorize dataset scoring or establish commercial, scientific or regulatory validation.